Confidential

Introduction

This room starts with a confidential case file linked to a group calling themselves black hat hackers. The task is to analyse the file carefully and find the secret invite code hidden within the evidence.

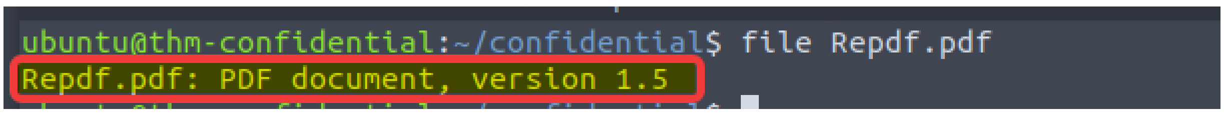

Figure 1: Identifying the downloaded case file as a PDF document.

The downloaded file was identified using the file command:

file Repdf.pdf

The output confirmed that Repdf.pdf is a PDF document using PDF version 1.5.

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Confidential File - Case[133713]

We are black hat hackers and we have created this secret QRcode to share the secret to all the blackhat hackers around the globe. We are redacting this information so that it doesn't get into the hands of those white hat hackers, we won't let them save the day again.

Here is redacted secret QR Code :

Keep it safe.

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Figure 2: Contents of the confidential PDF file showing the redacted QR code.

The PDF was opened to review its contents. It contained a confidential case file with a partially redacted QR code, described as a secret code intended for the group's members.

The PDF file was submitted to VirusTotal for further inspection. The analysis showed that none of the 63 security vendors flagged the file as malicious. VirusTotal also identified the uploaded document as newPDF.pdf and displayed its SHA-256 hash.

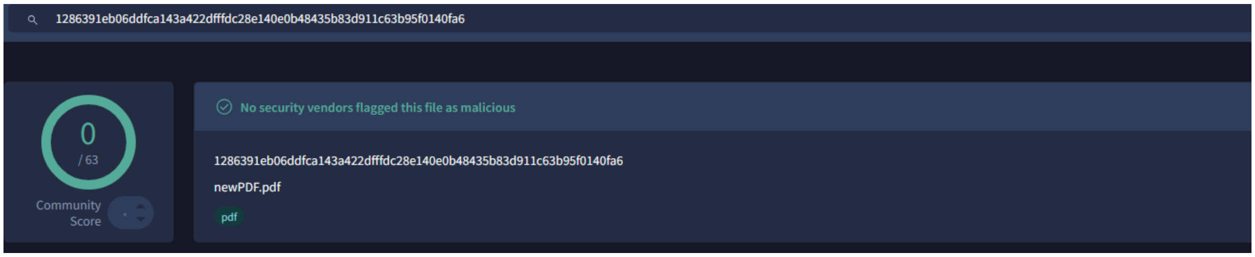

Figure 3: VirusTotal analysis showing that the PDF file was not detected as malicious.

Answer the questions below

Uncover and scan the QR code to retrieve the flag!

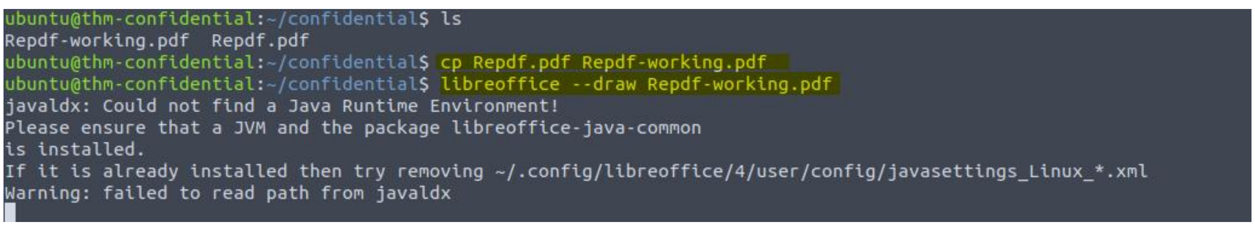

Figure 4: Creating a working copy of the PDF and opening it in LibreOffice Draw.

A working copy of the original PDF was created before making any changes:

cp Repdf.pdf Repdf-working.pdf

LibreOffice Draw was then launched with the copied file:

libreoffice --draw Repdf-working.pdf

The terminal displayed a warning that no Java Runtime Environment was available and suggested installing the libreoffice-java-common package. Despite this warning, LibreOffice Draw opened the PDF successfully, allowing the red exclamation-mark image covering the QR code to be moved or removed without altering the original evidence file.


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Figure 5: Moving the image that was covering the QR code in LibreOffice Draw.

The red exclamation-mark image was selected and moved away from the QR code. This revealed the complete QR code while keeping the edit limited to the working copy of the PDF.

Figure 6: The uncovered QR code.

The QR code was fully visible after the red warning image was moved away from it.

Decode Succeeded

Raw text	flag{e08e6ce2f077a1b420cfd4a5d1a57a8d}
Raw bytes	42 66 66 c6 16 77 b6 53 03 86 53 66 36 53 26 63 03 73 76 13 16 23 43 23 06 36 66 43 46 13 56 43 16 13 53 76 13 86 47 d0 ec 11 ec 11 ec 11 ec 11 ec 11 ec 11 ec 11 ec
Barcode format	QR_CODE
Parsed Result Type	TEXT
Parsed Result	flag{e08e6ce2f077a1b420cfd4a5d1a57a8d}

Figure 7: Successful QR-code decoding showing the recovered flag.

The recovered QR-code image was then uploaded to ZXing Decoder Online for decoding. The service identified the image as a QR code and successfully returned the hidden flag

The QR code decodes to:

Answer: flag{e08e6ce2f077a1b420cfd4a5d1a57a8d}

Conclusion

This was a short room focused on examining a PDF file and looking beyond what was immediately visible. The QR code had not been removed from the document; it was simply covered by a separate image. After creating a copy of the PDF, moving the image aside, and decoding the QR code, the hidden invite code was recovered.

Disclaimer

This writeup was created for educational and portfolio purposes only. All testing was performed in the relevant lab environment.